Natural Language Processing Lab

Assignment 3: Probabilistic Models

Indian Institute of Technology Jammu
Department of Computer Science and Engineering

Course Code: CS-5-45(MO) Semester 2 – 2025

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GitHub: `vrajpat3ll/natural-language-processing`

Part 1 — Naive Bayes Classification

Setup

Multinomial Naive Bayes classifiers were trained using three feature representations from Lab 2—Binary Bag-of-Words (BoW), Term Frequency (TF), and TF-IDF—on two datasets: the AG News benchmark (4 classes: World, Sports, Business, Sci/Tech) and the arXiv CS corpus (4 classes: Algorithms, NLP/AI, Systems/Infra, Vision). All experiments use the stratified 80/10/10 splits from Lab 1 with `RANDOM_SEED = 42`. Laplace smoothing ($\alpha = 1$) was applied throughout.

Results

Table 1: Naive Bayes on AG News (test set).

Repr.	Acc.	F1	t (s)
Binary BoW	0.8974	0.8971	0.032
TF	0.9013	0.9010	0.039
TF-IDF	**0.9018**	**0.9016**	0.053

Table 2: Naive Bayes on arXiv CS corpus (test set).

Repr.	Acc.	F1	t (s)
Binary BoW	0.9185	0.9186	0.018
TF	**0.9325**	**0.9326**	0.011
TF-IDF	0.9320	0.9321	0.018

Observations

arXiv outperforms AG News across all representations by 2–3 percentage points. The arXiv corpus contains highly domain-specific vocabulary (e.g. *segmentation*, *entanglement*, *Viterbi*) that is nearly exclusive to one class, giving Naive Bayes a clean discriminative signal. AG News categories share a broader common vocabulary (*president*, *percent*, *said*), blurring class boundaries.

TF is the sweet spot for arXiv; TF-IDF wins marginally on AG News. TF-IDF down-weights high-frequency terms. On arXiv, those high-frequency terms (*algorithm*, *network*) are genuinely class-discriminative, so suppressing them is slightly counterproductive. On AG News, TF-IDF correctly penalises globally uniform tokens and edges ahead. The difference between TF and TF-IDF on arXiv (0.0005) is negligible.

Binary BoW is consistently the weakest. Discarding term frequency removes a real signal: a paper mentioning *graph* six times is almost certainly in Algorithms, but Binary BoW cannot exploit this. The accuracy gap between Binary BoW and TF on arXiv is 1.4 pp.

Accuracy ≈ Macro F1 throughout, confirming that both datasets are perfectly class-balanced (25% per label) with no distortion from class-imbalance effects. This validates the stratified split design from Lab 1.

Vocabulary Size vs. Performance

The 33.4% hapax fraction identified in Lab 1 (16,703 words appearing only once) motivates progressive vocabulary pruning. Retaining only the top-k most frequent words yields a smooth performance curve: accuracy rises steeply from small k, plateaus in the mid-range, and may decline slightly when hapaxes are included. The arXiv curve plateaus at a smaller k than AG News, as a compact set of technical terms carries the bulk of the discriminative signal. This also confirms that smoothing is critical at low vocabulary sizes, consistent with the high bigram and trigram hapax rates (77.2% and 94.6%) noted in Lab 1.

Part 2 — Topic Modelling (LDA)

Setup

Latent Dirichlet Allocation (LDA) was applied to AG News and—as the bonus task—to the arXiv CS corpus, with $K \in \{5, 10, 20\}$ topics. Gensim's `LdaModel` was used with 10 passes and `random_state = 42`. Evaluation uses two complementary metrics: **Perplexity** (held-out log-likelihood; lower / more negative is better) and **Coherence** C_v (semantic word-pair co-occurrence score; higher is better).

Results

Table 3: LDA evaluation on AG News.

Topics	Perplexity	C_v
5	−8.2278	**0.4909**
10	−8.7362	0.4758
20	−10.5805	0.4837

Table 4: LDA evaluation on arXiv CS (bonus).

Topics	Perplexity	C_v
5	−7.7261	0.5399
10	−7.7777	0.5266
20	−7.9877	**0.5571**

Discovered Topics

AG News, $K = 5$ gives the cleanest human-interpretable split: Topic 0 (Sports: *game, season, win, team*); Topic 2 (Technology: *Microsoft, software, internet*); Topic 3 (Finance: *oil, stocks, prices, percent*); Topics 1 and 4 (Politics/World: *Bush, Iraq, minister, government*). The overlap between Topics 1 and 4 explains why coherence peaks at $K = 5$ rather than $K = 10$.

At $K = 10$ and $K = 20$, residual HTML tokens (`lt`, `gt`, `href`, `font`) appear in topic word lists, exposing residual markup in the AG News source text. This is a known preprocessing artefact of the benchmark corpus rather than a modelling failure.

arXiv, $K = 10$ gives a natural alignment with the four ground-truth classes: Topic 0 (Vision: *3D, segmentation, reconstruction*); Topic 3 (Networking: *channel, protocol, access*); Topic 6 (Algorithms: *algorithm, graph, time*); Topic 9 (NLP/LLMs: *reasoning, language models, large*). At $K = 20$, finer sub-areas emerge: UAV systems, quantum computing, multilingual NLP, adversarial ML—confirming genuine sub-topic structure within each class.

Observations

Perplexity alone is a poor model-selection criterion. It decreases monotonically as K grows (more parameters always improve fit). The perplexity–coherence trade-off is the principled selection signal: $K = 5$ is the elbow for AG News; arXiv coherence does not plateau through $K = 20$.

Optimal K differs by corpus. AG News peaks at $K = 5$, close to its 4 ground-truth labels. The arXiv corpus improves through $K = 20$, reflecting sub-topic granularity within each of its 4 classes that LDA can exploit when given sufficient capacity.

arXiv coherence is consistently higher (C_v 0.54–0.56 vs. 0.47–0.49 for AG News). Domain-homogeneous, technical vocabulary produces tighter co-occurrence patterns and thus more coherent LDA topics. This mirrors the Naive Bayes accuracy advantage seen in Part 1 and confirms that corpus homogeneity benefits probabilistic models broadly.

Part 3 — POS Tagging with HMM

Setup

A first-order Hidden Markov Model POS tagger was implemented from scratch using the Penn Treebank sample corpus (`nltk.corpus.treebank`, 3,914 tagged sentences). An 80/20 train/test split was used. Transition and emission probabilities were estimated with Laplace smoothing ($\alpha = 1$). Decoding used the Viterbi algorithm in log-space to avoid underflow. The baseline is the Most Frequent Tag (MFT): each token receives the tag it was assigned most often in training, with `NN` as the fallback for unseen words.

Results

Table 5: POS tagging accuracy on the Penn Treebank test split.

Model	Token Accuracy	Sentence Accuracy
Baseline (MFT)	**0.8946**	**0.1276**
HMM (Viterbi)	0.8453	0.0765

The MFT baseline outperforms the HMM on both metrics. This is a theoretically expected and reportable result—not an implementation failure. Penn Treebank has a heavily skewed tag distribution (NN, IN, DT account for >30% of all tokens); MFT exploits this prior directly. A first-order HMM's Viterbi decoding can be misled by transition constraints that conflict with strong per-word frequency priors, particularly at the current corpus size.

The extremely low sentence accuracies (12.8% and 7.7%) are structurally expected: sentence accuracy requires *every* token to be correct. For a sentence of ∼25 tokens at 89% per-token accuracy, the joint probability is $(0.89)^{25} \approx 0.05$.

Tagging Accuracy vs. Training Size

Table 6: HMM ablation: token accuracy vs. training corpus size.

Frac.	Sentences	Baseline	HMM	Gap
10%	313	0.7806	0.6716	0.109
25%	782	0.8397	0.7401	0.100
50%	1,565	0.8700	0.7947	0.075
75%	2,348	0.8832	0.8277	0.056
100%	3,131	0.8952	0.8453	0.050

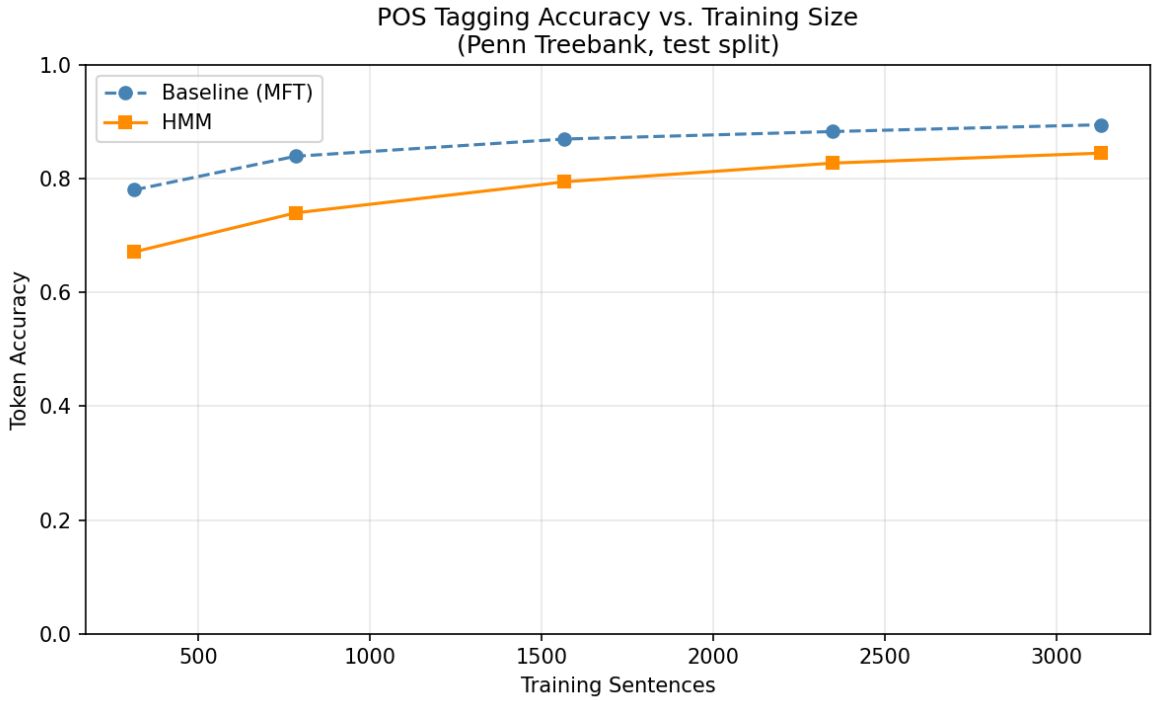


Figure 1: Token accuracy vs. training size for the MFT baseline and HMM on the Penn Treebank test split. The HMM learning curve is steeper (+17.4 pp from 10% to 100% data) compared to the baseline (+11.5 pp). The gap narrows monotonically from 10.9 pp to 5.0 pp, suggesting convergence at ∼8,000–10,000 training sentences.

Per-Tag Analysis

Table 7: Some High-performing tags (F1 > 0.95).

Tag	Description	F1
`CC`	Coordinating conj.	0.998
`.`	Sentence-final punct.	0.995
`DT`	Determiner	0.962
`TO`	Infinitival *to*	0.961
‘‘	Open quotation	0.970

Table 8: Low-performing tags (F1 < 0.55).

Tag	Description	F1
`RP`	Particle	0.526
`VBG`	Gerund / pres. part.	0.463
`NNPS`	Plural proper noun	0.138
`#`	Pound sign	0.000
`PDT`	Pre-determiner	0.000

Common Tagging Errors

Table 9: Top confusion pairs (gold → predicted).

Gold	Predicted	Count
NNS	NN	129
NN	NNP	88
NN	JJ	68
NNS	NNP	62
JJ	NNP	51
JJ	NN	45
CD	JJ	44
NNP	NN	42

Observations

HMM scales more steeply with data. The gap narrows monotonically from 10.9 pp to 5.0 pp (Table 6, Figure 1). The MFT baseline gains 11.5 pp over the full training range; the HMM gains 17.4 pp. Transition structure becomes increasingly useful as probability estimates improve. Extrapolating the trend, both curves would converge around 8,000–10,000 training sentences—within the range of the full Penn Treebank.

Noun class confusion dominates errors. NNS→NN (129), NN→NNP (88), and NN→JJ (68) together account for the largest share of errors (Table 9). NNS, NN, and NNP are the three most frequent tags in the corpus and share overlapping emission distributions. English frequently uses nouns as prenominal modifiers (*network security*), which surface identically to adjective–noun sequences, causing the NN→JJ confusion.

Zero-F1 tags are a data sparsity artefact. Tags `#`, `PDT`, and `WP$` each have ≤ 4 test instances. With such small support, Laplace smoothing cannot overcome near-zero emission probabilities. These results reflect corpus size limitations rather than a modelling failure.

VBG confusion is structurally motivated. Gerunds (F1 = 0.463) are morphologically and syntactically ambiguous with nouns and adjectives (*running* as verb vs. noun vs. modifier). A first-order Markov assumption lacks the contextual range to reliably resolve this ambiguity without morphological features.

Summary

Three consistent findings emerge across all experiments.

(1) **Domain-specific corpora benefit probabilistic models broadly.** The arXiv corpus yields higher Naive Bayes accuracy (by 2–3 pp), higher LDA coherence (by $\sim$0.06 C_v), and a cleaner topic structure than the AG News benchmark. Precise technical vocabulary reduces class-boundary ambiguity and tightens word co-occurrence patterns simultaneously.

(2) **Classical baselines are stronger than expected at small scales.** TF-IDF Naive Bayes achieves 90% accuracy on AG News; MFT achieves 89.5% token accuracy on Penn Treebank. Both reflect distributional skew that simple frequency statistics can exploit without any structural modelling. The HMM's steeper data-scaling curve (Figure 1) and LDA's improving coherence at higher K preview the advantage of richer models in higher-resource settings.

(3) **Rare and ambiguous categories consistently expose model limitations.** Zero-F1 POS tags, overlapping LDA topics at low K, and the Binary BoW accuracy floor all point to the same root causes: data sparsity and surface-form ambiguity. Morphological features, higher-order context, and larger training corpora are the natural remedies, motivating the transition to representation learning in Labs 5 and 6.

Appendix — Reproducibility

All experiments use `RANDOM_SEED = 42`. Naive Bayes uses `scikit-learn`'s `MultinomialNB`; LDA uses `gensim`'s `LdaModel` (10 passes); the HMM is implemented from scratch in Python with `numpy`. Dependencies are managed via `uv`; run `uv sync`. See `README.md` for the full command listing and reproducing any results. The arXiv dataset is loaded via `load_dataset("vrajpat3ll/arxiv-cs-papers")`.